Fractions and Percentages - Mark scheme

Number · Edexcel IGCSE Mathematics A (4MA1) · Year 10

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Question 1 *Practice question written for this worksheet (not from a past paper)*

Work out $\frac{2}{5}+\frac{1}{3}$

Common denominator 15: $\frac{6}{15}+\frac{5}{15}$

Answer: \dfrac{11}{15}

Question 2 *Practice question written for this worksheet (not from a past paper)*

Work out $\frac{3}{4}-\frac{2}{7}$

Common denominator 28: $\frac{21}{28}-\frac{8}{28}$

Answer: 13/28

Question 3 *Practice question written for this worksheet (not from a past paper)*

Work out $2\frac{1}{2}+1\frac{3}{4}$

Add whole numbers: 3; add fractions: $\frac{2}{4}+\frac{3}{4}=\frac{5}{4}$

$3+1\frac{1}{4}$

Answer: 4 1/4

Question 4 *Practice question written for this worksheet (not from a past paper)*

Work out $3\frac{1}{3}-1\frac{4}{5}$

Improper fractions: $\frac{10}{3}-\frac{9}{5}=\frac{50}{15}-\frac{27}{15}$

Answer: 23/15 = 1 8/15

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Question 5 *Practice question written for this worksheet (not from a past paper)*

Work out $\frac{2}{3} \times \frac{9}{10}$

$\frac{18}{30}$, divide top and bottom by 6

Answer: 3/5

Question 6 *Practice question written for this worksheet (not from a past paper)*

Work out $1\frac{1}{2} \times 2\frac{2}{3}$

$\frac{3}{2} \times \frac{8}{3} = \frac{24}{6}$

Answer: 4

Question 7 *Practice question written for this worksheet (not from a past paper)*

Work out $\frac{3}{5} \div \frac{9}{10}$

Flip and multiply: $\frac{3}{5} \times \frac{10}{9} = \frac{30}{45}$

Answer: 2/3

Question 8 *Practice question written for this worksheet (not from a past paper)*

Work out $2\frac{1}{4} \div 1\frac{1}{2}$. Give your answer as a mixed number.

$\frac{9}{4} \div \frac{3}{2} = \frac{9}{4} \times \frac{2}{3} = \frac{18}{12} = \frac{3}{2}$

Answer: 1 1/2

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Question 9 *Practice question written for this worksheet (not from a past paper)*

Pam spends $\frac{1}{3}$ of her money on rent and $\frac{1}{4}$ on food.

What fraction of her money is left?

Spent: $\frac{1}{3}+\frac{1}{4}=\frac{7}{12}$

Left: $1-\frac{7}{12}$

Answer: 5/12

Question 10 *Practice question written for this worksheet (not from a past paper)*

$\frac{3}{8}$ of a number is 24. Work out the number.

$\frac{1}{8}$ of the number is $24 \div 3 = 8$

Whole number: 8×8

Answer: 64

Question 11 *Practice question written for this worksheet (not from a past paper)*

In a class, $\frac{2}{5}$ of the students are boys.

$\frac{3}{4}$ of the boys wear glasses.

What fraction of the class are boys who wear glasses?

$\frac{2}{5} \times \frac{3}{4} = \frac{6}{20}$

Answer: 3/10

Question 12 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 4*

4	612 – 450 (= 162) **or** $\frac{612}{450}$ (= 1.36) oe		3	M1	for finding the difference or the fractional increase
	$\frac{"162"}{450}$ $\left(=\frac{9}{25}=0.36\right)$ or $\frac{"162"}{450} \times 100$ **or** "1.36" – 1 $\left(=\frac{9}{25}=0.36\right)$ or ("1.36" – 1) × 100 **or** "1.36" × 100 (=136) or ("1.36" × 100) – 100			M1	for a correct method to find the percentage increase or be one calculation away
	Correct answer scores full marks (unless from obvious incorrect working)	36		A1	
					Total 3 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 4

4	1 + 0.15 (= 1.15) **or** $x + 0.15x = 943$ **or** 100(%) + 15(%) (= 115(%)) **or** $\frac{943}{115}$ (= 8.2) oe		3	M1
	943 ÷ "1.15" **or** 943 ÷ "115" × 100 **or** 943 × 100 ÷ "115" oe **or** 8.2 × 100			M1 dep on M1
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	820		A1
				Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 5

5	eg 1 − 0.28 (= 0.72) oe **or** $0.72x = 198$ **or** 100(%) − 28(%) (= 72(%)) **or** $\frac{198}{72}$ (= 2.75) oe		3	M1 for a correct first step
	eg (x =) 198 ÷ "0.72" oe **or** 198 ÷ "72" × 100 oe **or** "2.75" × 100			M1 for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)	275		A1 cao
				Total 3 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 7

Question	Working	Answer	Mark	Notes	
7*	0.07 × 4000 (= 280) **or** 1.07 × 4000 (= 4280)		3	M1 for finding 7% of 4000 or 107% of 4000	M2 for $1.07^3 \times 4000$ or $1.07^4 \times 4000$ (= 5243….)
	4000 + "280" (= 4280) oe **and** 0.07 × "4280" (= 299.6) **and** "4280" + "299.6" (= 4579.6) **and** 0.07 × "4579.6" (= 320.572) **or** "280" + "299.6" + "320.572" (= 900(.172))			M1	
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	4900		A1 allow answers in the range 4900 – 4901 If no other mark awarded, SCB1 for 4000 × 0.07 × 3 (= 840) or 4000 × 0.21 (= 840) or 4000 + 4000 × 0.07 × 3 (= 4840) or 4000 × 1.21 (= 4840) or 0.93 × 4000 (= 3720) or 0.79 × 4000 (= 3160) or $0.93^3 \times 4000$ (= 3217…) or 4000 × 1.07^2 (= 4579…)	
				Total 3 marks	

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 8

8	100(%) + 17(%) (=117(%)) **or** 1 + 0.17 (=1.17) oe **or** $\frac{3276}{117}$ (= 28) oe		3	M1 may be seen embedded Do not allow (1 + 17%) unless correctly processed
	$\frac{3276}{"1.17"}$ **or** $\frac{3276}{117} \times 100$ oe			M1 for a complete method
	Correct answer scores full marks (unless from obvious incorrect working).	2800		A1
				Total 3 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 8

8	$1 - 0.17$ or 0.83 or $\frac{83}{100}$ **or** $100(\%) - 17(\%)$ or $83(\%)$ **or** $\frac{6225}{83}$ $(= 75)$ oe		3	M1
	$6225 \div$ "0.83" **or** $6225 \div$ "83" $\times 100$ **or** $6225 \times 100 \div$ "83" oe **or** 75×100			M1
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	7500		A1
				Total 3 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 8

8	18000×0.15 $(= 2700)$ oe **or** 18000×0.85 $(= 15\,300)$ oe		3	M1 for finding 15% or 85% of 18 000	M2 for 18000×0.85^4 oe **or** 18000×0.85^5 $(= 7986.(69\ldots))$ oe
	eg 18000×0.85^4 oe **or** "15300" $\times 0.85 \times 0.85 \times 0.85$ oe **or** "15300" $\times 0.85 (= 13005)$ oe **and** "13005" $\times 0.85 (= 11054.25)$ oe **and** "11054.25" $\times 0.85$ oe			M1 (dep) for a complete method	
		9396		A1 awrt 9396	
				If no marks awarded, award SCB1 for **or** 18000×0.85^2 $(= 13\,005)$ oe **or** 18000×0.85^3 $(= 11\,054.(25))$ oe **or** $18\,000 \times 0.4$ $(= 7200)$ oe **or** $18\,000 \times 1.15$ $(= 20700)$ oe **or** $18\,000 \times 1.15^4$ $(= 31482.(1125))$ oe	
					Total 3 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 9

9	$100(\%) + 12(\%)$ $(= 112(\%))$ **or** $1 + 0.12$ $(= 1.12)$ oe **or** $\frac{140}{112}$ $(=1.25)$ oe		3	M1	may be seen embedded. Do **not** allow $(1 + 12\%)$ unless correctly processed
	$\frac{140}{"1.12"}$ **or** $\frac{140}{112} \times 100$ oe			M1	for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)	125		A1	
					Total 3 marks

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 9

9	$1 - 0.15$ $(= 0.85)$ **or** $100(\%) - 15(\%)$ $(= 85(\%))$ **or** $\frac{612}{85}$ $(= 7.2)$ oe		3	M1	may be seen embedded Do not allow $(1 - 15\%)$ unless processed correctly
	$612 \div$ "0.85" oe **or** $612 \div$ "85" $\times 100$ oe **or** "7.2" $\times 100$			M1	for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)	720		A1	
					Total 3 marks

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 9

9	20 000 × 1.035 (= 20 700) oe or 20 000 × 0.035 (= 700) oe (NB: accept $\left(1+\frac{3.5}{100}\right)$ for 1.035 but not (1 + 3.5%))		3	M1	For finding 103.5% or 3.5% of 20 000	M2 for 20 000 × 1.035 [NB: $1.035^3 = 1.108717...$] **or** $20\,000 \times 1.035^4$ (= 22 950…)
	"20 700" × 1.035 (= 21 424.5) "21 424.5"× 1.035 oe eg $20700\times0.035=724.5$ & $20700+724.5=21424.5$ $21424.5\times0.035=749.85...$ & $21424.5+749.85...$			M1	dep for a complete method	
	Correct answer scores full marks (unless from obvious incorrect working)	22 174		A1	Allow 22 174 - 22175 (if you see the correct answer and then 20 000 is subtracted to give 2174 - 2175 then award full marks 2174 – 2175 with no working gains 2 marks) SCB2 for $\left(2000\times1.035^3\right)=2217...$ **[misread]** SCB2 for 22160 (20 000 × 1.108) SCB2 for 22180 (20 000 × 1.109) SCB1 if no marks awarded for any of these are seen (not necessarily the answer) $(20\,000 \times 1.035^n)$ 20 000 × 0.965 (= 17 972…….) 20 000 × 0.105 (= 2100) 20 000 × 1.105 ((= 22 100) 20 000 × 1.105 (= 21 424.5)	
						Total 3 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 9

9	4500 × 1.024 (= 4608) oe or 4500 × 0.024 (= 108)		3	M1		M2 for 4500×1.024^4 or 4500×1.024^5
	"4608" × 1.024 (= 4718.592) and "4718.592"× 1.024 (= 4831.838…) and "4831.838"× 1.024 (=4947.80…)			M1		
	Correct answer scores full marks (unless from obvious incorrect working)	4948		A1	4947 – 4948 if no other mark awarded, SCB1 for 4500 × 0.024 × 4 (= 432) 0.096 × 4500 (=432) or 4500 + 4500 × 0.024 × 4 (= 4932) 4500 × 1.096 (= 4932) 0.976 × 4500 (=4392) or 0.904 × 4500 (=4068) or $0.976^4 \times 4500$ (=4083…) or 4500×1.024^3 (= 4831.83...)	
						Total 3 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2026 · Question 10

Q	Working	Answer	Mark	Notes	
	200 000 × 0.014 (= 2800) oe **or** 200 000 × 1.014 (= 202 800) oe		3	M1 for finding 1.4% or 101.4% of 200 000	M2 for 200 000 × 1.014^3 oe **or** 200 000 × 1.014^4 (= 211 437...) oe
	eg 200 000 × 1.014^3 oe **or** "202 800" × 1.014 × 1.014 oe **or** "202 800" × 1.014 (=205 639.2) oe **and** "205 639.2" × 1.014 oe			M1 (dep) for a complete method	
	Correct answer scores full marks (unless from obvious incorrect working)	208 518		A1 awrt 208 518 If no other mark awarded, award SCB1 for 200 000 × 0.042 (= 8400) oe **or** 200 000 × 1.042 (= 208 400) oe **or** 200 000 × 1.014^2 (= 205 639.2) oe **or** 200 000 × 0.986 (= 197 200) oe **or** 200 000 × 0.958 (= 191 600) oe **or** 200 000 × 0.986^3 (= 191 717...) oe	
					Total 3 marks

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 3

3	1500 + (36 × 450) (= 17 700)		4	M1
	"17 700" – 12 500 (= 5200) oe or $\frac{\text{"17 700"}}{12\,500}$ (=1.416) oe			M1
	$\frac{\text{"5200"}}{12500}$ (× 100) oe or 0.416 (× 100) or $\frac{\text{"17 700"}}{12\,500}$ × 100 (– 100) or 141.6 (– 100)			M1
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	41.6		A1 allow 42 from correct working
				Total 4 marks

Question 25

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2023 · Question 3

3	$0.7 \times 60 \times 22\ (=924)$ oe **or** $(1-0.7) \times 60 \times 19\ (=342)$ oe **OR** $0.7\times60\times\left(22-\frac{780}{60}\right)(=378)$ oe **or** $(1-0.7)\times60\times\left(19-\frac{780}{60}\right)(=108)$ oe		4	M1	for finding income for the 22 dirhams notebooks **or** the 19 dirhams notebooks **OR** for finding the profit for the 22 dirhams notebooks **or** the 19 dirhams notebooks
	$0.7 \times 60 \times 22\ (=924)$ oe **and** $(1-0.7) \times 60 \times 19\ (=342)$ oe **OR** $0.7\times60\times\left(22-\frac{780}{60}\right)(=378)$ oe **and** $(1-0.7)\times60\times\left(19-\frac{780}{60}\right)(=108)$ oe			M1	for finding income for the 22 dirhams notebooks **and** the 19 dirhams notebooks **OR** for finding the profit for the 22 dirhams notebooks **and** the 19 dirhams notebooks, 1266 **or** 486 implies M2
	eg $\frac{"924"+"342"-780}{780}\times100$ **or** $\frac{"924"+"342"}{780}\times100-100$ **or** $\frac{"378"+"108"}{780}\times100$ **or** $\frac{486}{780}\times100$			M1	for a complete method to find percentage profit
	Correct answer scores full marks (unless from obvious incorrect working)	62.3		A1	awrt 62.3, allow 62 from correct working
					Total 4 marks

Question 26

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2023 · Question 4

4	$1600 \times 0.16\ (=256)$ oe or $1-0.16\ (=0.84)$ oe		4	M1	
	$1600 - "256"$ or $1600 \times "0.84"\ (=1344)$			M1	
	$\frac{"1344"}{1400}(=0.96)$ **or** $\frac{1400-"1344"}{1400}(=0.04)$ **or** $\frac{"1344"}{1400}\times100(=96)$ **or** $\frac{1400-"1344"}{1400}\times100$			M1	
	Correct answer scores full marks (unless from obvious incorrect working)	4		A1	SCB1 for 1856 seen if no other marks awarded
					Total 4 marks

Question 27

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2023 · Question 6

6	eg $1-\frac{1}{6}$ or $\frac{5}{6}$ $1-0.16(666\ldots)\ (=0.83(333\ldots))$ oe or $100(\%)-16(.666\ldots)(\%)\ (=83(.333\ldots)(\%))$ oe or $1-0.2\ (=0.8)$ oe or $100(\%)-20(\%)\ (=80(\%))$		4	M1 Allow eg $1-0.16\ (=0.84)\ (=84(\%))$ $1-0.17\ (=0.83)\ (=83(\%))$ rounded or truncated
	$140, "\frac{5}{6}"\ (=168)$ oe or $140 \div "0.83(333\ldots)"\ (=168)$ oe eg $140 \div 83.33 \times 100\ (=168)$ **or** $136 \div "0.8"\ (=170)$ oe eg $136 \div 80 \times 100\ (=170)$			M1 Allow eg $140 \div "0.84" = 166(.666\ldots)$ $140 \div "0.83" = 168(.674\ldots)$ rounded or truncated
	$140, "\frac{5}{6}"\ (=168)$ oe or $140 \div "0.83(333\ldots)"\ (=168)$ oe **and** $136 \div "0.8"\ (=170)$ oe			M1 Allow eg $140 \div "0.84" = 166(.666\ldots)$ $140 \div "0.83" = 168(.674\ldots)$ rounded or truncated
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	2		A1 Allow –2
				Total 4 marks

Question 28

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2024 · Question 7

7	475 × 0.16 (= 76) oe **or** 475 × (1 – 0.16) (= 399) oe		4	M1 (working for shop A)
	1 – 0.15 (= 0.85) **or** $x - 0.15x = 408$ **or** 100(%) – 15(%) (= 85(%)) **or** $\frac{408}{85}$ (= 4.8) oe			M1 (working for shop B)
	408 ÷ "0.85" (= 480) **or** 408 ÷ "85" × 100 (= 480) **or** 408 × 100 ÷ "85" (= 480) oe **or** "4.8" × 100 (= 480) **or** $\frac{408}{85} \times 15 (= 72)$			M1 (working for shop B)
	Working required	**A** **and** 72 and 76 seen		A1 dep on M2 for **A** with correct working (72 and 76 seen)
				Total 4 marks

Question 29

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 8

Question	Working		Answer	Mark	Notes
8	$\frac{2.9}{100} \times 5000 (= 145)$ oe **or** 1.029 × 5000 (=5145) oe **or** $1.029^2 \times 5000$ (= 5294….) oe **or** 0.058 × 5000 (= 290) oe **or** 1.058 × 5000 (= 5290)				M1 **Bank H**
	5000 × 0.016 oe (= 80) oe **or** 5000 × 1.016 oe (= 5080) oe **or** 5000 × 0.032 (= 160) oe **or** 5000 × 1.032 (= 5160) oe	M2 for 5000×1.016^2 (= 5161.28)		4	M1 **Bank G**
	(80 + 5000) × 0.016 (= 81.28) oe **or** 5080 × 1.016 (= 5161.28) oe				M1 **Bank G**
	Correct answer scores full marks (unless from obvious incorrect working)		16.28		A1
					Total 4 marks

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Question 30

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 10

10	$0.15 \times 32\,000\ (=4800)$ oe **or** $(1-0.15) \times 32\,000\ (=27\,200)$ oe		4	M1	for a correct method to find 15% **or** 85% of 32 000 allow this mark for sight of 36 800 or 48 668 or 14 400 or 17 600	M2	for $0.85^3 \times 32\,000\ (=19\,652)$ or $0.85^4 \times 32\,000\ (=16\,704)$
	eg $0.15 \times (32\,000 - \text{“}4800\text{”}) (=4080)$ **and** $0.15 \times (32\,000 - \text{“}4800\text{”} - \text{“}4080\text{”}) (=3468)$ **and** $32\,000 - (4800 + 4080 + 3468) (=19\,652)$ **OR** $(1-0.15) \times \text{“}27\,200\text{”} (=23\,120)$ oe **and** $(1-0.15) \times \text{“}23\,120\text{”} (=19\,652)$ oe **OR** $0.85^2 \times \text{“}27\,200\text{”} (=19\,652)$			M1	for a complete method to find the value of the car after 3 years Do **not** accept $(1-15\%)$ in place of $(1-0.15)$ unless correctly processed		
	$\frac{\text{“}19\,652\text{”}-2516}{3\times12}$ or $17\,136 \div (3 \times 12)$ oe or $\frac{[\text{value of the car}]-2516}{3\times12}$ oe			M1	dep on M1 for a method to find the monthly payment [value of the car] is what they believe to be the value of the car after 3 years. There must be some attempt to use **more than one year** of depreciation and the value must satisfy $2516 < [\text{value of the car}] < 32\,000$		
	Correct answer scores full marks (unless from obvious incorrect working)	476		A1	cao SCB3 for either of the following: • answer in the range [394, 395] (use of 4 years) • answer 1282 (use of 15% increase) SCB2 for: • answer 419 (simple depreciation)		
					Total 4 marks		

Question 31

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2018 · Question 13

13	$\frac{1}{3}+\frac{1}{5}\ (=\frac{8}{15})$ or 0.53... or 53.3.....% or 53%	900	4	M1
	$1-\text{“}\frac{8}{15}\text{”}\ (=\frac{7}{15})$ or 0.46..... or 0.47 or 46.6...% or 47%			M1
	$420 \div \text{“}\frac{7}{15}\text{”}\ (=900)$ oe			M1
				A1
				Total 4 marks

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Question 32

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 4

4		$45 \div 1.5\ (=30)$ **or** $5 \times 12\ (=60)$ **or** $\frac{64}{1.5}\ \left(=\frac{128}{3}=42.6(6...)\right)$		5	M1	for a method to find the number of boxes needed **or** the cost of adhesive **or** cost of tiles per m^2
		"30" × 64 (= 1920) **or** "42.6(6…)" × 45 (=1920)			M1	for a method to find the cost of the boxes of tiles
		"1920" + "60" (= 1980) **or** 3000 – 1920 – 60 (=1020)			M1	for a method to find the total cost **or** the profit
		eg $\frac{3000-"1980"}{"1980"}\ (=0.515...)$ **or** $\frac{3000-"1980"}{"1980"}\times 100$ **or** $\frac{3000}{"1980"}\ (=1.515\ldots)$ **or** $\frac{3000}{"1980"}'\ 100\ (=151.5\ldots)$ **or** $\frac{3000}{"1980"}\times 100-100$			M1	for a method to find the percentage profit or be one step away
		Correct answer scores full marks (unless from obvious incorrect working)	51.5		A1	awrt 51.5 SCB3 for answer 56.3 or 56.25 (from use of 1920 instead of 1980 as total cost)
						Total 5 marks

Question 33

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 4

4	(a)	$357 \div 0.17$ oe or $0.17N = 357$ or $\frac{17}{100}\times N = 357$ oe or $\frac{357\times 100}{17}$ oe eg $357 \times 5.8(82\ldots)$		2	M1	a correct calculation for N or a correct equation in N (**not** $17\% \times N = 357$)
		Correct answer scores full marks (unless from obvious incorrect working)	2100		A1	cao
	(b)	$806 - 650\ (=156)$ or $\frac{806}{650}(=1.24)$ oe		3	M1	
		$\frac{806-650}{650}(\times 100)(=0.24(\times 100))$ or "1.24" ×100 (= 124) or "1.24" – 1 (= 0.24)			M1	a correct calculation for the percentage increase or seeing 124 or 0.24 as either the answer or in part of the working.
		Correct answer scores full marks (unless from obvious incorrect working)	24		A1	cao (SCB1 if no marks scored for an answer of 19.3 – 19.4)
						Total 5 marks

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Question 34

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 13

13	$\frac{3}{8}+45\%\left(=\frac{33}{40}\text{ or }82.5(\%)\text{or }0.825\right)$		5	M1 Do **NOT** award M1 for e.g. $\frac{3}{8}+45(\%)+406(=....)$ oe
	$1-\frac{"33"}{40}\left(=\frac{7}{40}\right)$ **or** $100-"82.5"(\%)\left(=17.5(\%)\right)$ **or** $1-"0.825"(=0.175)$			M1
	$406\div\frac{"7"}{40}(=2320)$ **or** $406\div\frac{"17.5"}{100}$ oe $(=2320)$ **or** 1% = 406 ÷ "17.5" (= 23.2) oe			M1
	0.45 × "2320" oe **or** 45 × "23.2" oe			M1
		1044		A1
				Total 5 marks

13 ALT	$\frac{3}{8}x+0.45x+406$ oe		5	M1 Do **NOT** award M1 for e.g. $\frac{3}{8}+45(\%)+406(=....)$ oe
	$\frac{3}{8}x+0.45x+406=x$ oe			M1 for a correct equation
	$(x=)\frac{406}{1-\frac{3}{8}-0.45}\left(=\frac{406}{7/40}=2320\right)$			M1
	0.45 × "2320"			M1
		1044		A1
				Total 5 marks

Question 35

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 14

14	1 + 1.45 (= 2.45) or $1+\frac{29}{20}(=\frac{49}{20})$ or B = 1.45A oe or $B=\frac{29}{20}$A oe or A + 1.45A or $A+\frac{29}{20}$A or 2.45A or (A : B =) 100 : 145 oe or 100 + 145 (= 245) oe or (B : C =) 3 : 2 oe or $B = 1.5C$ oe			5	B1 must identify ratios with Abel and Bahira or Bahira and Chanda Allow any letters for A, B and C
	A + 1.45A = 15 435 or 15 435 ÷ "2.45" or 15 435 ÷ "$\frac{49}{20}$" or 15 435 ÷ "245" × 100 or 63 × 100 (= 6300)	M2 for $15\,435\div\left(\frac{1}{1.45}+1\right)(=9135)$ oe or $15\,435\div\left(\frac{49}{29}\right)(=9135)$ oe			M1 for a method to find Abel's savings or for 6300
	15 435 – "6300" or 1.45 × "6300" or 145 × "63" (= 9135)				M1 for a method to find Bahira's savings or for 9135
	"9135" ÷ $\frac{3}{2}$ oe or "9135" × $\frac{2}{3}$ oe				M1 for a method to find to find Chanda's savings
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)		6090		A1
					Total 5 marks

Question 36

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 5

5	(a)	$87 - 75\ (= 12)$ or $\frac{87}{75}(=1.16)$ oe		3	M1
		$\frac{"12"}{75}(\times 100)(=0.16(\times 100))$ oe or $\frac{87}{75}\times 100(=116)$ oe or $"1.16"\times 100(=116)$ oe			M1
		Correct answer scores full marks (unless from obvious incorrect working)	16		A1
	(b)	$0.2 \times 960\ (= 192)$ oe or $0.8 \times 960\ (= 768)$ oe or $0.8 \times 0.8 \times 960\ (= 614.4)$ oe or $0.8^3\ (= 0.512)$ oe		3	M1 NB Use of % sign to indicate the calculation, eg 80% × 960, is **not** sufficient for this mark unless an answer of 768 is stated for this calculation
		$0.8^3 \times 960$ oe or $0.8 \times 0.8 \times$ "768" oe or $0.8 \times$ "614.4" oe or $0.2 \times$ "768" $(= 153.6)$ **and** $0.2 \times$ ("768" − "153.6") $(= 122.88)$ oe or "192" + "153.6" + "122.88" $(= 468.48)$			M1 NB Use of % sign to indicate the calculation, eg $(1 - 20\%)^3 \times 960$, is **not** sufficient for this mark unless an answer of 491.52 is stated for this calculation
		Correct answer scores full marks (unless from obvious incorrect working)	491.52		A1 491 - 492 SC B2 for $0.8^4 \times 960\ (= 393(.216))$ If no other mark awarded, SCB1 for $960 \times 0.2 \times 3\ (= 576)$ oe or $960 \times 0.4\ (= 384)$ oe or $960 + 960 \times 0.2 \times 3\ (= 1536)$ oe or $960 \times 1.6\ (= 1536)$ oe or $1.2 \times 960\ (= 1152)$ oe or $1.2^2 \times 960\ (= 1382(.4))$ oe or $1.2^3 \times 960\ (= 1658(.88))$ oe
					Total 6 marks

Question 37

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 9

9	(a)	for 0.04×2000 oe $(= 80)$ **or** 1.04×2000 oe $(= 2080)$	**OR**		3	M1	for finding 4% **or** 104% of 2000	**OR** M2 for 2000×1.04^3 oe **or** 2000×1.04^4 oe $(= 2339.72)$
		$1.04 \times$ "2080" oe $(= 2163.2)$ $1.04 \times$ "2163.2" oe	2000×1.04^3 oe			M1	for completing method to find total amount in the account at the end of 3 years	
		Correct answer scores full marks (unless from obvious incorrect working)		2250		A1	accept 2249 – 2250	
							SC: if no other marks gained award M1 for 0.12×2000 oe **or** 240 **or** 1.12×2000 oe **or** 2240 accept (1 + 0.04) as equivalent to 1.04 throughout	
	(b)	eg $1365 \div (1 - 0.09)$ **or** $1365 \div 0.91$			3	M2	for a complete method	
						(M1)	for $1365 \div (100 - 9)\ (= 15)$ **or** $(100 - 9)\% = 1365$ **or** $91\% = 1365$ **or** eg $(1 - 0.09)T = 1365$ **or** eg $T - 0.09T = 1365$	
		Correct answer scores full marks (unless from obvious incorrect working)		1500		A1		
								Total 6 marks

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Question 38 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 6*

6	(a)	$7 \times 5 \times 14$ (= 490) oe **or** 7×14 (= 98) **and** $400 \div 5$ (= 80)		4	M1 for working out the pay per week or pay per day
		"490" – 400 (= 90) **or** "98" – "80" (= 18) **or** "98" ÷ "80" oe **or** "490" ÷ 400 oe **or** 1.225 oe			M1
		$\frac{"90"}{400}(\times 100)(=0.225)$ oe **or** $\frac{"18"}{"80"}(\times 100)(=0.225)$ oe **or** $\frac{"98"}{"80"} \times 100 (=122.5)$ oe **or** $\frac{"490"}{400} \times 100 (=122.5)$ oe **or** "1.225" – 1 (= 0.225)			M1 dep on M2
		Correct answer scores full marks (unless from obvious incorrect working)	22.5		A1 oe allow 23% with M3 awarded
	(b)	E.g. 1 − 0.06 (= 0.94) or 100(%) − 6(%) (= 94(%)) or $\frac{23\,030}{94}$ (= 245) oe		3	M1
		E.g. 23 030 ÷ "0.94" or 23 030 ÷ "94" × 100 or 23 030 × 100 ÷ "94" or "245" × 100			M1
		Correct answer scores full marks (unless from obvious incorrect working)	24 500		A1
					Total 7 marks

CHALLENGE QUESTIONS

Question 39 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 6*

6	(a)	18 000 + 14 × 1160 (= 34 240) oe or 18 000 + 16 240 (= 34 240)		4	M1
		"34 240" – 32 000 (= 2240) or $\frac{"34\,240"}{32\,000}$ (= 1.07)			M1
		$\frac{"2240"}{32\,000}(\times 100)$ or $\frac{"34\,240"}{32\,000} \times 100$ (= 107) or "1.07" – 1 (= 0.07)			M1
		Correct answer scores full marks (unless from obvious incorrect working)	7		A1
	(b)	e.g. 1 − 0.15 (= 0.85) or 100(%) − 15(%) (= 85(%))		3	M1
		e.g. 39 865 ÷ 0.85 or 39 865 ÷ 85 × 100 oe			M1
		Correct answer scores full marks (unless from obvious incorrect working)	46 900		A1
					Total 7 marks

Question 40

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 12

12		$16\,000 \times \left(1-\frac{12}{100}\right)^2$ (= 12 390.4) **or** 16 000 × 0.7744 (= 12 390.4) oe **or** $16\,000 \times \left(1-\frac{12}{100}\right)$ (= 14 080) **and** "14 080" × $\left(1-\frac{12}{100}\right)$ (=12 390.4) oe **or** $\left(1-\frac{12}{100}\right)^2$ (=0.7744) **and** $\frac{11461.12}{16000}$ $\left(=\frac{4477}{6250}=0.71632\right)$		3	M1	for a method to find the value of the car after two years **or** a method to find the overall percentage multiplier after two years **and** the overall percentage multiplier for the three years May be seen embedded, eg in an equation Do not allow (1 – 12%) unless processed correctly
		eg $\frac{\text{"12390.4"}-11461.12}{\text{"12390.4"}}$ (×100) (= 0.075) **or** $1-\frac{11461.12}{\text{"12390.4"}}$ (×100) **or** 1 – 0.925 (= 0.075) **or** $\frac{11461.12}{\text{"12390.4"}}$ (×100) (= 0.925) **or** $\frac{\text{"0.71632"}}{\text{"0.7744"}}$ (′ 100) (= 0.925)			M1	for a complete method to find the value of the decimal equivalent of x% **or** for a complete method to find the percentage multiplier for the third year May be seen embedded, eg within a correct equation rearranged to one of these equivalent forms
		Correct answer only scores full marks (unless from obviously incorrect working)	7.5		A1	oe SCB2 for answer –7.5
						Total 3 marks

Question 41

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 13

13	750 000 × (1 – 0.04) (= 720 000) oe **and** "720 000" × (1 – 0.065) (= 673 200) **or** 750 000 × (1 – 0.04) × (1 – 0.065) (= 673 200) **or** 750 000 × 0.96 × 0.935 (= 673 200)		3	M1	(NB: accept $\left(1-\frac{4}{100}\right)$ for 0.96 but not (1 – 4%) and accept $\left(1-\frac{6.5}{100}\right)$ for 0.935 but not (1 – 6.5%)) Calculations may be seen as part of an equation eg 750 000 × 0.96 × 0.935 × $\left(1+\frac{x}{100}\right)$ = 698 445 1.0375 or 25245 imply M1
	eg $\frac{698\,445-\text{"673 200"}}{\text{"673 200"}}$ (×100) (= 0.0375) **or** $\frac{698\,445}{\text{"673 200"}}-1$ (= 0.0375) **or** 1.0375 – 1 (= 0.0375) oe **or** $\frac{698\,445}{\text{"673 200"}}\times 100\,(=103.75)$			M1	for a method to reach value one step away from x ie a method leading to 0.0375 or 103.75
	Correct answer scores full marks (unless from obvious incorrect working)	3.75		A1	oe eg $3\frac{3}{4}$ or $\frac{15}{4}$
					Total 3 marks

Question 42

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 13

13	$80\,000\times\left(\frac{100+x}{100}\right)^3=80\,000+6151.25$ oe **or** $80\,000\times\left(1+\frac{x}{100}\right)^3=80\,000+6151.25$ oe **or** $80\,000\times(1+x\%)^3=80\,000+6151.25$ oe **or** $80\,000\times y^3=80\,000+6151.25$ oe **or** $\frac{80\,000+6151.25}{80\,000}(=1.076\ldots)$ oe **or** $\frac{86\,151.25}{80\,000}(=1.076\ldots)$ oe		5	M1
	$\sqrt[3]{\frac{80\,000+6151.25}{80\,000}}(=1.025)$ oe **or** $\sqrt[3]{"1.076..."}(=1.025)$ **or** $\left(1+\frac{x}{100}=\right)\frac{41}{40}(=1.025)$			M1
	Correct answer scores full marks (unless from obvious incorrect working)	2.5		A1 Accept answers in the range 2.4 – 2.6 from correct working NB Do not allow an answer in the range 2.4 – 2.6 if it comes from awrt 7.6% oe or 7.7% oe divided by 3 Do not accept an answer if it is in the range that comes from a simple interest method
				Total 3 marks

Question 43

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 14

Question		Working	Answer	Mark	Notes
		1.2 or 0.9 or $1.2n$ **or** $0.9n$ where n is a number or a variable eg 1.2×100 **or** 0.9×100 **or** $1.2n\times\left(\frac{100-x}{100}\right)=0.9n$		3	M1 for one correct multiplier, may be seen as part of a single term or calculation or part of an equation
		$0.9\div1.2\ (=0.75)$ oe or 75(%) $\frac{n\times0.9}{n\times1.2}$ where n is a number or variable **or** $1-(0.9\div1.2)\ (=0.25)$ **or** $\frac{100-x}{100}=\frac{0.9n}{1.2n}$ oe			M1 for a correct step to find the percentage decrease for October **or** for an answer of 75(%) or 0.25 this mark implies the first M1
		Correct answer scores full marks (unless from obvious incorrect working)	25		A1
					Total 3 marks

Question 44

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2021 · Question 14

14	$0.85 \times x^2 = 1.0285$ **or** $85 \times x^2 = 102.85$ oe or $(x^2 =)\ 1.0285 \div 0.85$ **or** $(x^2 =)\ 102.85 \div 85$ oe **or** 1.21 oe		4	M2	for a correct equation using their chosen letter or value in place of letter, or a correct division or 1.21 seen otherwise: (M1 for either 0.85 or 1.0285 seen)
	$(x =) \sqrt{1.0285 \div 0.85}$ **or** $(x =) \sqrt{102.85 \div 85}$ oe **or** $(x =)$ 1.1(0)			M1	for a correct expression or value for x
		10		A1	
					Total 4 marks

Alternative Mark Scheme for Q14					
14	$\left(\frac{100+y}{100}\right)^2 \times 0.85 = 1.0285$ oe **or** $\left(\frac{100+y}{100}\right)^2 = 1.21$ oe or $10^4 + 200y + y^2 = 12100$ oe		4	M2	for a correct equation using their chosen letter, otherwise: (M1 for either 0.85 or 1.0285 seen)
	$\frac{100+y}{100} = 1.1$ **or** $100 + y = 110$ oe or $(y + 210)(y - 10) = 0$			M1	for a correct equation involving y with no square terms **or** a correct method for solving the quadratic: If factorising, allow brackets which expanded give 2 out of 3 terms correct – if using formula or completing the square allow one sign error and some simplification – allow as far as eg $\frac{-200 \pm \sqrt{40000 + 8400}}{2}$ or eg $(y + 100)^2 - 12100 = 0$ oe
		10		A1	
					Total 4 marks

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